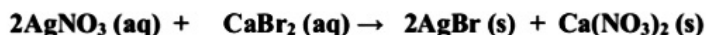


SUGGESTED ANSWER PSPM 1
SESI 2017/2018

1. a) A 25.0 mL of 0.050 M silver nitrate solution is mixed with 25.0 mL of 0.050 M of calcium bromide solution to give 0.105 g of solid silver bromide.

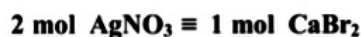
- i. Write the balanced chemical equation for this reaction.



- ii. Determine the limiting reactant.

$$\begin{aligned} n \text{ AgNO}_3 &= n \text{ CaBr}_2 = \frac{0.05 \times 25}{1000} \\ &= 1.25 \times 10^{-3} \text{ mol} \end{aligned}$$

From equation,



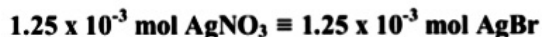
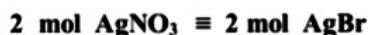
$$\begin{aligned} 1.25 \times 10^{-3} \text{ mol AgNO}_3 &\equiv \frac{1.25 \times 10^{-3} \text{ mol AgNO}_3 \times 1 \text{ mol CaBr}_2}{2 \text{ mol AgNO}_3} \\ &= 6.25 \times 10^{-4} \text{ mol CaBr}_2 \end{aligned}$$

Number of mole of CaBr_2 needed ($6.25 \times 10^{-4} \text{ mol}$) < number of mol CaBr_2 available ($1.25 \times 10^{-3} \text{ mol}$)

CaBr_2 is excess reactant and limiting reactant is AgNO_3

- iii. Calculate the percentage yield of silver bromide.

From equation,



$$\begin{aligned} \text{mass AgBr} &= 1.25 \times 10^{-3} \text{ mol} \times 187.8 \text{ g mol}^{-1} \\ &= 0.2348 \text{ g} \end{aligned}$$

$$\begin{aligned} \% \text{ yield} &= \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 \\ &= \frac{0.105}{0.2348} \times 100 \\ &= 44.72\% \end{aligned}$$

- b) An organic compound contains only carbon, hydrogen and oxygen with a mass composition of 41.40% C, 3.47% H and 55.13% O. If 0.05 mol of this compound weighs 5.80 g, determine its

- i. empirical formula

	C	H	O
% mass	41.40	3.47	55.13
Moles	$\frac{41.40}{12} = 3.45$	$\frac{3.47}{1} = 3.47$	$\frac{55.13}{16} = 3.45$
ratio	1	1	1

Empirical formula is CHO

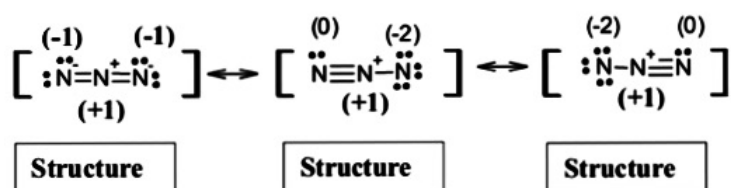
- ii. molecular formula

$$\begin{aligned}
 (\text{CHO})_n &= \text{molar mass} \\
 (12n + n + 16n) &= \frac{5.80 \text{ g}}{0.05 \text{ mol}} \\
 29n &= 116 \\
 n &= 4
 \end{aligned}$$

Molecular formula is C₄H₄O₄

2. a) The azide ion, N₃⁻, exists in several resonance forms.

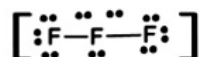
- i. Draw possible resonance structures for the azide ion.



- ii. Use formal charges to select the most stable structure.

Structure 1.

- iii. Using Lewis structure, explain why N_3^- ion exists, whereas trifluoride ion, F_3^- , does not exist.



F is in period 2, central atom cannot form expanded octet, so

F_3^- ion does not exist.

N is in period 2, central atom obey octet rule, so N_3^- ion is exists.

[9 marks]

- b) Chloroform, CHCl_3 , is a common organic solvent. If H in CHCl_3 is replaced by Cl, it becomes CCl_4 , a toxic solvent. For each of CHCl_3 and CCl_4 compound.

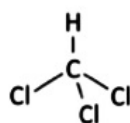
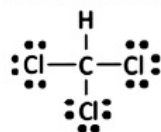
- i. draw the molecular shape.



No. of valence electron :

C	=	4
H	=	1
3 Cl	=	21
Total	=	26

Lewis structure :



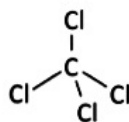
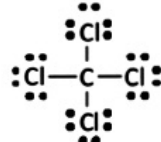
molecular shape: tetrahedral



No. of valence electron :

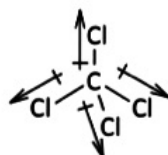
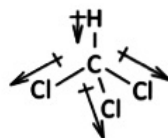
C	=	4
4 Cl	=	28
Total	=	32

Lewis structure :



molecular geometry : tetrahedral

- ii. show the bond polarity,



- iii. predict the polarity.

The magnitude of C-H dipole moment is not equal to C-Cl dipole moment. The four dipole moments do not cancel out each other, $\mu \neq 0$. Therefore CHCl3 is an asymmetry and a polar molecule.

All C-Cl dipole moments are equal in magnitude. The four dipole moments are arranged in tetrahedral shape causes them to cancel out each other, $\mu = 0$. Therefore CCl4 molecule is a symmetry and a non-polar molecule.

3. A 3.0 L flask at 298 K contains a mixture of 2.0 mol helium gas and 3.0 mol xenon gas. The van der Waals constants for helium and xenon are given in TABLE 3.

TABLE 3

Gas	$a \text{ (L}^2 \text{ atm mol}^{-2}\text{)}$	$b \text{ (L mol}^{-1}\text{)}$
Helium	0.03421	0.02370
Xenon	4.194	0.05105

Given the van der Waals equation: $\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$

- a) State **two (2)** postulates of the kinetic molecular theory of an ideal gas in relationship to parameters a and b as in TABLE 3.

Kinetic molecular theory of an ideal gas in relationship to:

parameter a -The attractive and repulsive forces between gas molecules are negligible.

@

The intermolecular forces are negligible.

parameter b -The combined volume of all molecules of the gas is negligible relative to volume of the gas container.

@

The total volume of all gas molecules is negligible compared to the volume in which the gas is contained.

- b) Which gas is expected to behave ideally and which gas exhibits a marked deviation from ideal behavior?

Helium is expected to behave ideally while xenon exhibits a marked deviation from ideal behaviour.

- c) Explain your answer in 3(b) by referring to the significance of a and b values in TABLE 3.

Helium has smaller value of a which means the attractive and repulsive forces between helium atoms are very weak and almost negligible. Helium also has a smaller value of b which means the combined volume of all helium atoms is negligible compared to the volume of the gas container. Therefore, helium behaves almost like an ideal gas.

Whereas, xenon has a bigger value of a which means the attractive and repulsive forces between xenon atoms are stronger and significant. A bigger value of b means that the combined volume of all xenon atoms is significant compared to the volume of the gas container. Therefore, xenon deviates from ideal gas behaviour.

- d) Calculate total pressure of the gas in the mixture using an ideal gas equation.

$$\begin{aligned} P_T &= \frac{n_T RT}{V} \\ &= \frac{(2.0 + 3.0) \text{ mol} \times 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 298 \text{ K}}{3.0 \text{ L}} \\ &= 41 \text{ atm} \end{aligned}$$

@

$$\begin{aligned} P_{\text{He}} &= \frac{n_{\text{He}} RT}{V} \\ &= \frac{2.0 \text{ mol} \times 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 298 \text{ K}}{3.0 \text{ L}} \\ &= 16.3 \text{ atm} \end{aligned}$$

$$\begin{aligned} P_{\text{Xe}} &= \frac{n_{\text{Xe}} RT}{V} \\ &= \frac{3.0 \text{ mol} \times 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 298 \text{ K}}{3.0 \text{ L}} \\ &= 24.5 \text{ atm} \end{aligned}$$

$$P_T = 16.3 \text{ atm} + 24.5 \text{ atm}$$

$$= 41 \text{ atm}$$

- e) Using the van der Waals equation, the total pressure of the system is calculated to be 38.1 atm. Explain the reasons behind the differences between this value and the pressure calculated in 3(d).

Pressure of a gas is the result of the collisions between the gas molecules with the wall of their container. The higher the frequency and impact of the collisions, the higher the gas pressure.

Helium and xenon are real gases with weak attractive forces between atoms. These forces lessen the impact of collision of a given atom with the wall of the container. Therefore, the actual pressure calculated using the van der Waals equation is lower than the pressure calculated using the ideal gas equation.

The combined volume of gas molecules is NOT the reason in this case because bigger gas molecules reduces the unoccupied volume and therefore increases the gas pressure.

4. a) In a series of acids of the halogen group, hydroiodic acid, HI, is the strongest acid while hydrofluoric acid, HF, is the weakest.
- i. Explain what is meant by strong and weak acids.

Strong acid is an acid that dissociates completely, while weak acid is an acid that dissociates partially.

- ii. Briefly explain the main factor in determining the strength of the acid.

The polarity of the bond formed between atom F and H. The weaker the bond, the easily it can dissociates to produce high concentration of H_3O^+ .

- iii. Calculate the degree of dissociation, α of 0.20 M HF solution.
[$K_a = 6.8 \times 10^{-4}$ at 25°C]

[] / M	$\text{HF (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{F}^- \text{ (aq)} + \text{H}_3\text{O}^+ \text{ (aq)}$		
I	0.20	-	-
C	-x	+x	+x
E	$0.20 - x$	x	x

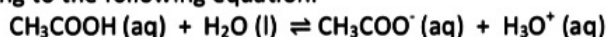
$$K_a = \frac{[\text{H}_3\text{O}^+][\text{F}^-]}{[\text{HF}]}$$

$$6.8 \times 10^{-4} = \frac{(x)^2}{(0.20 - x)}$$

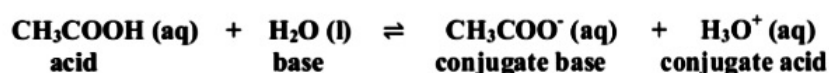
$$x_1 = 0.01133, x_2 = -0.01200 \text{ (rejected)}$$

$$\begin{aligned}\text{degree of dissociation, } \alpha &= \frac{x}{0.20} \\ &= \frac{0.01133}{0.20} \\ &= 0.057\end{aligned}$$

- b) Acetic acid, CH_3COOH , is a weak acid with $K_a = 1.8 \times 10^{-5}$. It dissociates in water according to the following equation:



- i. Rewrite the above equation and label the conjugate acid and conjugate base.



- ii. Determine the pH of 0.10 M.

[] / M	$\text{CH}_3\text{COOH (aq)}$	$+$	$\text{H}_2\text{O (l)}$	$\rightleftharpoons$	$\text{CH}_3\text{COO}^- \text{ (aq)}$	$+$	$\text{H}_3\text{O}^+ \text{ (aq)}$
I	0.10				0		0
C	-x				+x		+x
E	0.10 - x				x		x

$$\begin{aligned}K_a &= \frac{[\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}]} \\ 1.8 \times 10^{-5} &= \frac{(x)^2}{(0.10-x)}\end{aligned}$$

Since K_a is very small, assume $0.10 - x \approx 0.10$

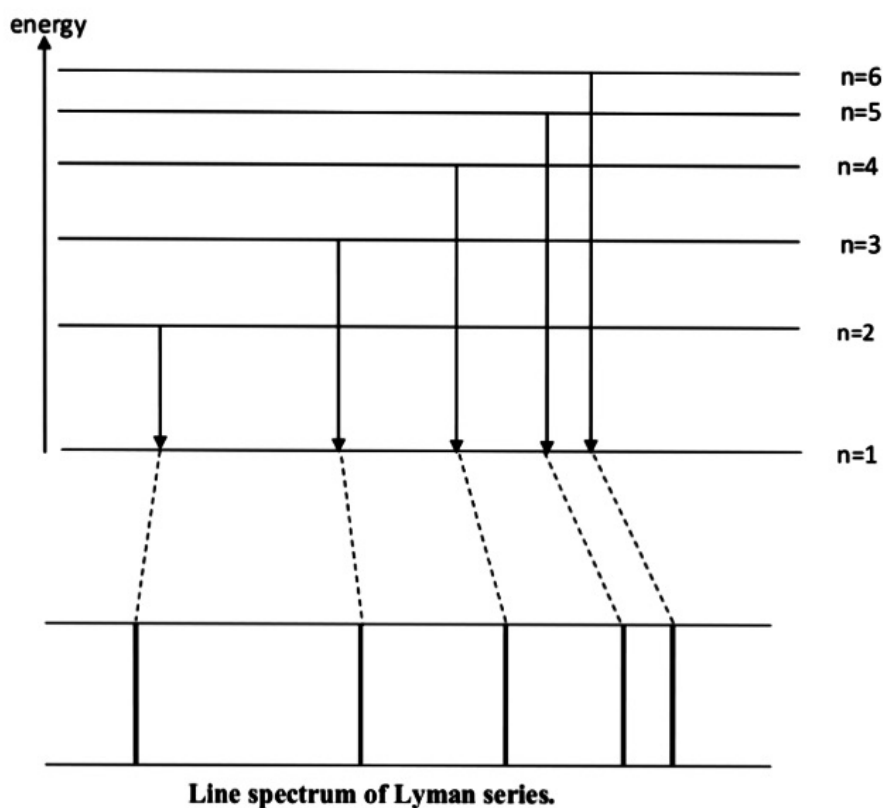
$$\begin{aligned}1.8 \times 10^{-5} &= \frac{(x)^2}{0.10} \\ x &= 1.34 \times 10^{-3} \text{ M} = [\text{H}_3\text{O}^+]\end{aligned}$$

$$\begin{aligned}\text{pH} &= -\log [\text{H}_3\text{O}^+] \\ &= -\log 1.34 \times 10^{-3} \\ &= 2.87\end{aligned}$$

- iii. Predict the percentage of dissociation when the acid concentration is increased.

Percent of dissociation remain unchanged when the acid concentration is increased.

5. a) Sketch the energy level diagram to show the electronic transitions which give rise to the first five lines in the Lyman series of the hydrogen atom. Explain.



When energy is supplied to an atom, electron at the ground state will absorb the energy. This electron will excite to higher energy level and the electron is now at its excited state. The electron is unstable and will fall back to lower energy level, $n = 1$ by releasing specific amount of energy with specific wavelength in the form of photon. The photon released is passed through a prism and fall on the photograph plate to be recorded as line spectrum.

5. b) Element Y is in Period 3 of the Periodic Table. The first six successive ionization energies of element Y are given in Table 5.

TABLE 5

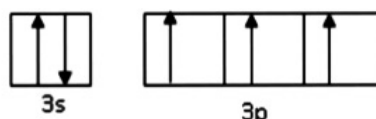
Ionization energy (kJ mol^{-1})	First	Second	Third	Fourth	Fifth	Sixth
	1011	1907	2914	4963	6274	21267

Determine the group for Y and explain your answer. Write the valence electronic configuration and the valence orbital diagram of Y. Explain how the Pauli exclusion

principle, Aufbau principle and Hund's rule are applied in drawing the valence orbital diagram of element Y.

- Sharp increase occur at IE_6 ($IE_6 \gg IE_5$).
- The sixth electron is removed from inner shell.
- Element Y has 5 valence electron.
- Element Y in group 15
- Valence electronic configuration of Y : $3s^2 3p^3$

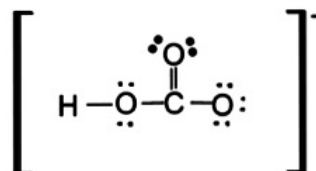
Valence orbital diagram :



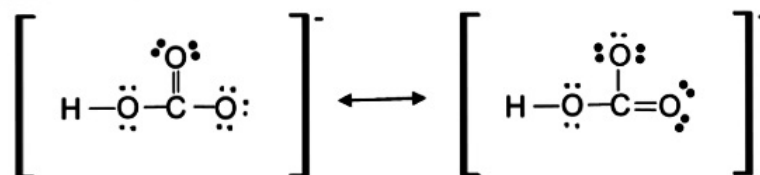
- Pauli exclusion principle state that no two electrons can have same set of four quantum number. Hence, only 2 electrons are located at same orientation with different spin.
- Aufbau principle state that in the ground state of an atom or ion, electrons fill atomic orbitals of the lowest energy level before higher energy level. Energy level 3s orbital is lower than 3p orbital.
- Hund's rule state that when electron are filled on degenerate orbital (such as 3p orbital), electron are filled singly with same spin before its paired.

6. a) Sodium bicarbonate, $NaHCO_3$ is used in baking as raising agent. When dissolved in water, the salt dissociates into ions. Write the Lewis structure for the bicarbonate ion, $HOCO_2^-$, and draw its resonance structures.

Lewis structure:



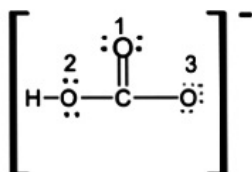
Resonance Structure:



Determine the shape of the $HOCO_2^-$ ion and the hybridization of central atoms. Draw and label the overlapping of the orbitals to show the formation of the covalent bonds.

Molecular Geometry for the $HOCO_2^-$ ion = Trigonal planar

Type of hybridisation:

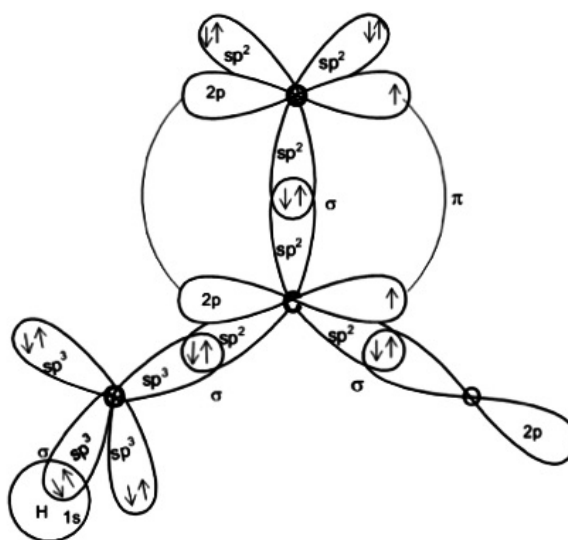


- i. $\text{O}_1 = \text{sp}^2$
- ii. $\text{O}_2 = \text{sp}^3$
- iii. $\text{C} = \text{sp}^2$

Valence Orbital diagram:

$\text{H} = \begin{array}{ c } \hline \uparrow \\ \hline 1s \\ \hline \end{array}$ $\text{O}_3 = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow & \uparrow \\ \hline 2s & 2p & & \end{array}$	$\text{O}_2 =$ $\text{GS} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow \\ \hline 2s & 2p & & \end{array}$ $\text{ES} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow \\ \hline \text{sp}^2 & & 2p & \end{array}$
$\text{O}_1 =$ $\text{GS} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow \\ \hline 2s & 2p & & \end{array}$ $\text{ES} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow \\ \hline 2s & 2p & & \end{array}$ $\text{ES} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow \\ \hline \text{sp}^3 & & & \end{array}$	$\text{C} =$ $\text{GS} = \begin{array}{ c c c c } \hline \uparrow\downarrow & \uparrow & \uparrow & \uparrow \\ \hline 2s & 2p & & \end{array}$ $\text{ES} = \begin{array}{ c c c c } \hline \uparrow & \uparrow & \uparrow & \uparrow \\ \hline 2s & 2p & & \end{array}$ $\text{ES} = \begin{array}{ c c c c } \hline \uparrow & \uparrow & \uparrow & \uparrow \\ \hline \text{sp}^2 & & 2p & \end{array}$

Orbital Overlapping:



- (b) Metallic compounds have some physical properties which are different from covalent compounds. State four (4) properties and relate these properties to metallic bonding.

[8 marks]

Four properties of metallic compound:

- i) **High melting and boiling point**
Metallic compounds have strong electrostatic attraction between positively charged metal ion and negatively charge electron sea. More energy needed to break this attraction.
- ii) **Able to conduct electricity**
Metallic compounds have free moving electrons.
- iii) **Malleable**
Delocalized electrons enable the metal atoms to roll over each other. Metal atoms only slide to each other.
- iv) **Luster**
Photons of light do not penetrate very far into the surface of a metal and are typically reflected on the metallic surface.

- 7 (a) Carbon dioxide can be used to extract caffeine under supercritical condition. Some of the physical properties of carbon dioxide are shown below.

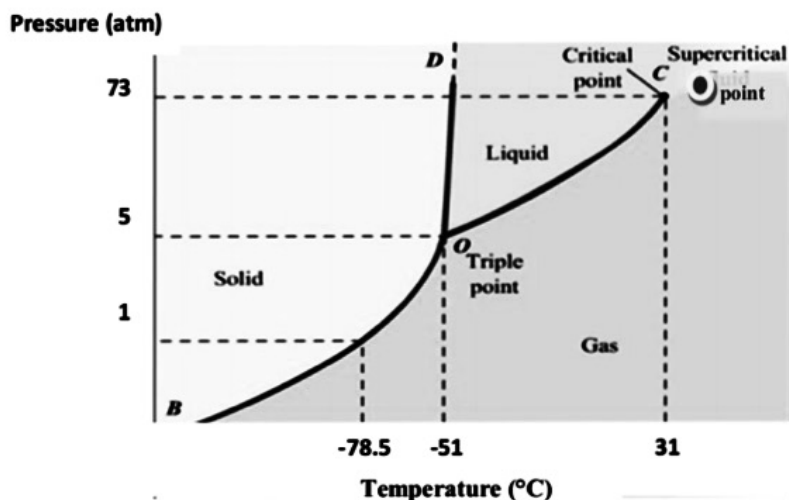
Triple point temperature	-51°C
Triple point pressure	5 atm
Critical temperature	31°C
Critical pressure	73 atm

What is meant by triple point and critical point?

Triple point:	Temperature and pressure at which solid, liquid and gas simultaneously exist in equilibrium.
Critical point:	Temperature and pressure at which gaseous phase and liquid phase are indistinguishable.

Using the data given above, sketch a labeled phase diagram for carbon dioxide. Show the supercritical point in the diagram.

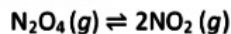
Phase diagram of carbon dioxide



Using the phase diagram, under what conditions can dry ice be formed to act as coolants?

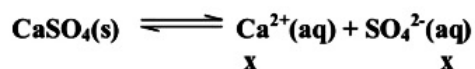
Dry ice can be formed to act as coolants under atmospheric pressure (1 atm) and the temperature of -78.5°C . Under these conditions, dry ice breaks down and turns directly into carbon dioxide gas rather than a liquid. The super-cold temperature and the sublimation feature make dry ice great for refrigeration or as coolants. For example, if we want to send something frozen across the country, we can pack it in dry ice. It will be frozen when it reaches its destination, and there will be no messy liquid left over like we would have with normal ice.

- b) In an experiment, 0.10 mol of N_2O_4 and 0.55 mol of NO_2 are mixed in a closed vessel at 350 K with a total pressure of 2.0 atm. At this temperature, the equilibrium constant, K_p is 3.89. The reaction is given as follows:



In which direction will the reaction proceed to reach equilibrium?

- b) The solubility product, K_{sp} for calcium sulphate, CaSO_4 , is 2.0×10^{-5} at 25°C . Calculate the molar solubility of CaSO_4 in water and in 0.10 M sodium sulphate, Na_2SO_4 . State common ion effect on the solubility of CaSO_4 .

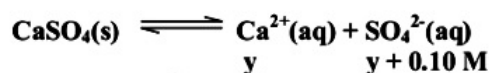


$$K_{sp} = [\text{Ca}^{2+}][\text{SO}_4^{2-}]$$

$$2 \times 10^{-5} = x^2$$

$$x = 4.47 \times 10^{-3} \text{ M}$$

Molar solubility in water is $4.47 \times 10^{-3} \text{ M}$



$$K_{sp} = [\text{Ca}^{2+}][\text{SO}_4^{2-}]$$

$$2 \times 10^{-5} = (y)(y + 0.10) \quad ; \text{ assume } x \ll 0.10, y + 0.10 = 0.10$$

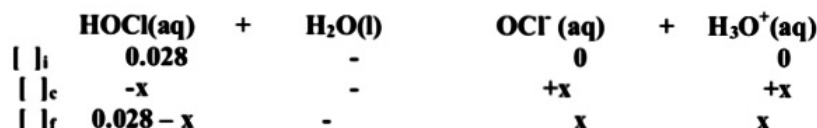
$$2 \times 10^{-5} = (y)(0.10)$$

$$y = 2 \times 10^{-4} \text{ M}$$

Molar solubility in Na_2SO_4 is $2 \times 10^{-4} \text{ M}$

- Solubility of CaSO_4 in pure water higher than in 0.10 M Na_2SO_4 solution, SO_4^{2-} (common ion) is present.
- The equilibrium position shifts backward.
- Solubility of CaSO_4 decreases.
- The presence of common ion, SO_4^{2-} reduces the solubility of CaSO_4 .

8. a) Hypochlorous acid, HOCl, is a monoprotic acid. An aqueous solution of 0.028 M HOCl has a pH of 4.5. Calculate the K_a value for this acid. A buffer solution is prepared by mixing 100 mL of 0.04 M HOCl with 100 mL of 0.02 M NaOCl. Calculate the pH of the solution. Explain how a mixture of HOCl and sodium hypochlorite, NaOCl, solution behaves as a buffer solution when a small amount of strong acid is added.



$$\begin{aligned} \text{pH} &= -\log [\text{H}_3\text{O}^+] \\ 4.5 &= -\log x \\ x &= 3.16 \times 10^{-5} \text{ M} \end{aligned}$$

$$\begin{aligned} [\text{HOCl}] &= 0.02797 \text{ M} \\ [\text{OCl}^-] &= [\text{H}_3\text{O}^+] = 3.16 \times 10^{-5} \text{ M} \end{aligned}$$

$$\begin{aligned} K_a &= \frac{[\text{OCl}^-][\text{H}_3\text{O}^+]}{[\text{HOCl}]} \\ &= \frac{(3.16 \times 10^{-5})(3.16 \times 10^{-5})}{0.02797} \\ &= 3.57 \times 10^{-8} \end{aligned}$$

$$n \text{ HOCl} = (0.04)(0.1) = 4 \times 10^{-3} \text{ mol}$$

$$[\text{HOCl}] = \frac{4 \times 10^{-3}}{0.2} = 0.02 \text{ M}$$

$$n \text{ NaOCl} = (0.02)(0.1) = 2 \times 10^{-3} \text{ mol}$$

$$[\text{NaOCl}] = \frac{2 \times 10^{-3}}{0.2} = 0.01 \text{ M}$$

$$\begin{aligned} \text{pH} &= -\log K_a + \log \frac{[\text{OCl}^-]}{[\text{HOCl}]} \\ &= -\log 3.57 \times 10^{-8} + \log \frac{[0.01]}{[0.02]} \\ &= 7.15 \end{aligned}$$

Buffer solution contain HOCl act as an acid and OCl⁻ that acts as a base. When small amount of strong acid is added to it, OCl⁻ will neutralized it. Thus, the pH is not much affected.



$$\begin{aligned}
 P_{\text{N}_2\text{O}_4} &= X_{\text{N}_2\text{O}_4} \times P_{\text{total}} \\
 &= \frac{n_{\text{N}_2\text{O}_4}}{n_{\text{total}}} \times P_{\text{total}} \\
 &= \frac{(0.1 \text{ mol})}{(0.10 + 0.55) \text{ mol}} \times 2.0 \text{ atm} \\
 &= 0.3077 \text{ atm}
 \end{aligned}$$

$$\begin{aligned}
 P_{\text{NO}_2} &= X_{\text{NO}_2} \times P_{\text{total}} \\
 &= (1 - X_{\text{N}_2\text{O}_4}) \times P_{\text{total}} \\
 &= (1 - 0.1538) \times 2.0 \text{ atm} \\
 &= 1.6923 \text{ atm}
 \end{aligned}$$

$$Q_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})} = \frac{(1.6923)^2}{(0.3077)} = 9.31$$

Since $Q_p > K_p$:

- The system is not at equilibrium
- Initially, there are more NO_2 in the reaction mixture
- To reach equilibrium $Q_p = K_p$, the reaction will shift backward

If the above experiment is carried out at 500 K, the new K_p is 1700. Is the reaction exothermic or endothermic? Explain.

- At higher temperature, the value of K_p increases
- The system consumes more N_2O_4 and produces more NO_2
- This indicates that at higher temperature, the equilibrium system is disturbed
- According to Le Chatelier's principle, equilibrium position will be shifted to the right as to lower the temperature of the system by absorbing the added heat
- Therefore, the forward reaction is endothermic